

## BALIKPAPAN SEPINGGAN, Indonesia

WMO: 966330

Lat: 1.2625S

Lon: 116.8947E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |       |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-------|------|------|-------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |       |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR    | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)   | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 73.9       | 74.8 | 70.9                          | 114.0 | 79.3 | 71.6 | 116.9 | 79.3 | 18.8                  | 84.0 | 16.2 | 85.0 | 5.8                      | 0    | 0.369 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 9.5                    | 91.3            | 81.1 | 89.9 | 80.6 | 89.2 | 80.4 | 82.7                | 88.3 | 81.9 | 87.5 | 81.4 | 86.9 | 8.7                  | 70   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.0                            | 161.0 | 86.4 | 80.5 | 158.5 | 86.1 | 79.9 | 155.0 | 85.4 | 46.7          | 88.8 | 45.9 | 87.7 | 45.2 | 87.7 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
| (a)               | (b)  | (c)  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |
| 17.1              | 14.5 | 12.8 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
|                   |      |      | 70.2                       | 94.5 | 4.3                | 2.5 | 67.1                                               | 96.3 | 64.6       | 97.7 | 62.2       | 99.1 | 59.0       | 100.9 |
|                   |      |      | 68.5                       | 84.1 | 3.9                | 1.1 | 65.7                                               | 84.8 | 63.5       | 85.5 | 61.3       | 86.1 | 58.5       | 86.8  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |       |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 81.9   | 81.5  | 81.7  | 82.0  | 82.0  | 82.5  | 81.7  | 81.2  | 81.6  | 82.3  | 82.5  | 82.3  | 81.9 |
|                                                                               | DBStd       | 1.82   | 1.93  | 1.88  | 1.77  | 1.79  | 1.79  | 1.79  | 1.73  | 1.68  | 1.68  | 1.85  | 1.72  | 1.74 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 11653  | 975   | 888   | 993   | 961   | 1006  | 951   | 966   | 981   | 968   | 1006  | 968   | 990  |
|                                                                               | CDD65       | 6178   | 510   | 468   | 528   | 511   | 541   | 501   | 501   | 516   | 518   | 541   | 518   | 525  |
|                                                                               | CDH74       | 66040  | 5024  | 4743  | 5505  | 5428  | 6095  | 5456  | 5281  | 5570  | 5845  | 6133  | 5585  | 5375 |
|                                                                               | CDH80       | 20676  | 1513  | 1506  | 1781  | 1711  | 2056  | 1641  | 1423  | 1550  | 1869  | 2100  | 1833  | 1693 |
| Wind                                                                          | WSAvg       | 5.9    | 5.3   | 6.0   | 5.5   | 5.1   | 5.3   | 5.8   | 6.6   | 7.5   | 7.2   | 5.9   | 4.9   | 5.2  |
| Precipitation                                                                 | PrecAvg     | 106.5  | 10.5  | 8.8   | 10.0  | 9.2   | 10.1  | 9.4   | 9.4   | 9.0   | 7.0   | 7.0   | 7.6   | 9.4  |
|                                                                               | PrecMax     | 135.5  | 28.7  | 14.9  | 17.8  | 21.1  | 21.7  | 17.9  | 19.3  | 29.9  | 22.1  | 15.9  | 15.0  | 16.2 |
|                                                                               | PrecMin     | 78.5   | 2.2   | 0.4   | 3.9   | 4.6   | 3.5   | 2.5   | 1.4   | 0.3   | 0.0   | 0.8   | 1.3   | 4.7  |
|                                                                               | PrecStd     | 13.7   | 5.6   | 3.8   | 3.7   | 3.7   | 4.1   | 4.3   | 4.6   | 6.5   | 4.8   | 3.9   | 3.9   | 3.1  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 91.4  | 91.7  | 91.1  | 91.2  | 91.6  | 91.0  | 88.8  | 89.8  | 90.1  | 91.3  | 91.5  | 91.6 |
|                                                                               |             | MCWB   | 79.6  | 80.1  | 80.4  | 81.7  | 82.6  | 81.9  | 80.4  | 80.6  | 81.0  | 81.4  | 81.4  | 80.8 |
|                                                                               | 2%          | DB     | 89.4  | 89.8  | 89.5  | 89.4  | 89.7  | 88.2  | 87.5  | 87.8  | 88.2  | 89.4  | 89.6  | 89.6 |
|                                                                               |             | MCWB   | 79.7  | 79.6  | 80.1  | 80.8  | 81.4  | 80.7  | 80.1  | 80.1  | 80.1  | 80.5  | 80.8  | 80.3 |
|                                                                               | 5%          | DB     | 87.9  | 88.3  | 88.1  | 87.9  | 88.1  | 87.0  | 86.2  | 86.4  | 87.4  | 87.9  | 88.2  | 88.1 |
|                                                                               |             | MCWB   | 79.6  | 79.5  | 79.9  | 80.4  | 80.9  | 80.3  | 79.5  | 79.3  | 79.7  | 80.0  | 80.0  | 79.8 |
|                                                                               | 10%         | DB     | 86.4  | 87.0  | 86.9  | 86.6  | 86.9  | 85.9  | 85.2  | 85.5  | 86.2  | 86.8  | 87.1  | 86.8 |
|                                                                               |             | MCWB   | 79.0  | 79.2  | 79.6  | 79.8  | 80.4  | 79.8  | 79.1  | 78.8  | 79.1  | 79.6  | 79.6  | 79.3 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 81.9  | 82.4  | 82.7  | 83.2  | 83.6  | 82.8  | 81.8  | 81.7  | 82.0  | 82.6  | 82.7  | 82.4 |
|                                                                               |             | MCDB   | 87.7  | 89.0  | 88.3  | 88.8  | 89.2  | 87.9  | 86.1  | 87.1  | 88.2  | 88.7  | 89.1  | 88.9 |
|                                                                               | 2%          | WB     | 80.9  | 81.0  | 81.6  | 81.8  | 82.5  | 81.7  | 80.8  | 80.6  | 80.9  | 81.3  | 81.6  | 81.1 |
|                                                                               |             | MCDB   | 86.8  | 87.1  | 87.1  | 87.2  | 87.7  | 86.6  | 85.4  | 85.9  | 86.6  | 87.2  | 87.7  | 87.3 |
|                                                                               | 5%          | WB     | 80.1  | 80.3  | 80.7  | 80.9  | 81.6  | 80.8  | 80.1  | 79.9  | 80.2  | 80.6  | 80.8  | 80.4 |
|                                                                               |             | MCDB   | 85.8  | 86.0  | 86.1  | 86.2  | 86.6  | 85.6  | 84.7  | 84.9  | 85.5  | 86.4  | 86.6  | 86.2 |
|                                                                               | 10%         | WB     | 79.5  | 79.6  | 80.0  | 80.2  | 80.8  | 80.1  | 79.5  | 79.2  | 79.7  | 80.0  | 80.0  | 79.7 |
|                                                                               |             | MCDB   | 84.8  | 85.1  | 85.2  | 85.2  | 85.6  | 84.6  | 83.9  | 84.0  | 84.8  | 85.4  | 85.5  | 85.1 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.8  | 11.1  | 10.8  | 10.0  | 9.7   | 9.0   | 8.7   | 8.5   | 8.8   | 9.5   | 10.5  | 10.9 |
|                                                                               |             | MCDBR  | 12.7  | 12.8  | 12.2  | 11.6  | 11.5  | 10.6  | 10.0  | 9.5   | 10.1  | 10.9  | 12.1  | 12.8 |
|                                                                               |             | MCWBR  | 5.6   | 5.6   | 5.6   | 5.5   | 5.5   | 5.4   | 5.1   | 4.7   | 4.8   | 5.1   | 5.5   | 5.8  |
|                                                                               | 5% WB       | MCDBR  | 11.7  | 11.9  | 11.4  | 10.8  | 10.7  | 9.8   | 9.5   | 9.2   | 9.6   | 10.1  | 11.5  | 11.8 |
| MCWBR                                                                         |             | 5.8    | 5.8   | 5.9   | 5.6   | 5.6   | 5.4   | 5.2   | 4.9   | 4.8   | 5.1   | 5.6   | 5.8   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.405  | 0.416 | 0.417 | 0.410 | 0.408 | 0.411 | 0.419 | 0.468 | 0.506 | 0.538 | 0.459 | 0.412 |      |
|                                                                               | taud        | 2.498  | 2.458 | 2.453 | 2.478 | 2.501 | 2.503 | 2.477 | 2.310 | 2.186 | 2.082 | 2.330 | 2.482 |      |
|                                                                               | Ebn at Noon | 294    | 291   | 288   | 283   | 276   | 271   | 271   | 263   | 259   | 254   | 276   | 290   |      |
|                                                                               | Edh at Noon | 36     | 38    | 38    | 36    | 34    | 33    | 34    | 42    | 49    | 54    | 42    | 36    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1410   | 1546  | 1556  | 1531  | 1483  | 1395  | 1488  | 1573  | 1608  | 1551  | 1451  | 1422  |      |
|                                                                               | RadStd      | 87     | 146   | 120   | 92    | 68    | 77    | 139   | 143   | 125   | 107   | 64    | 97    |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page